

# Mikroişlemci Sistemleri

#9\_2 – Kesmeler 2

YTÜ-CE

# Ders-11 Konular

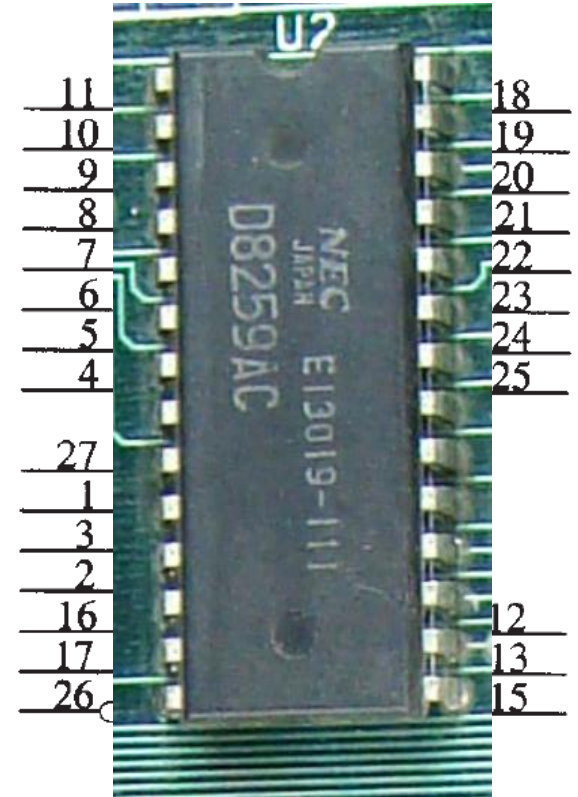
- 8259A
  - Uç Tanımları
  - İç Yapı
  - Ayarlama
    - ICWs
    - OCWs
- 8259A Örneği

# 8259A Programmable Interrupt Controller

- 8259A programlanabilir kesme kontrol devresidir.
- Kesme isteklerine öncelik atayabilir (priority encoding)
- Tek başına 8 farklı kesme vektörü sağlar
- Master (1)/Slave (8) formunda 64 farklı kesme vektörü sağlar

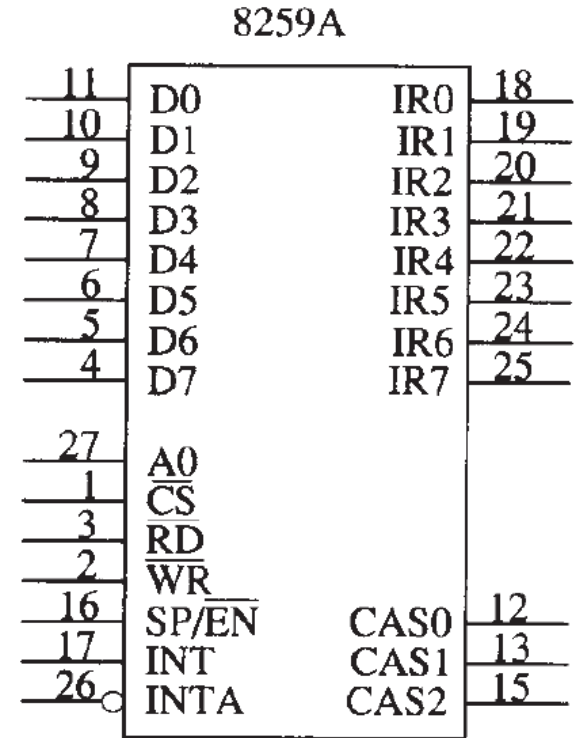
# 8259A PIC

- D0-D7: Data pinleri
- IR0-IR7: Kesme istek pinleri
- $\overline{WR}$ ,  $\overline{RD}$ ,  $\overline{CS}$
- INT:  $\mu$ P INTR ucuna
- $\overline{INTA}$ :  $\mu$ P  $\overline{INTA}$  ucundan
- A0: Adres ucu
- CAS2-CAS0: Kaskat seçim uçları
- $SP/\overline{EN}$ : Slave program/enable buffer



# 8259A PIC

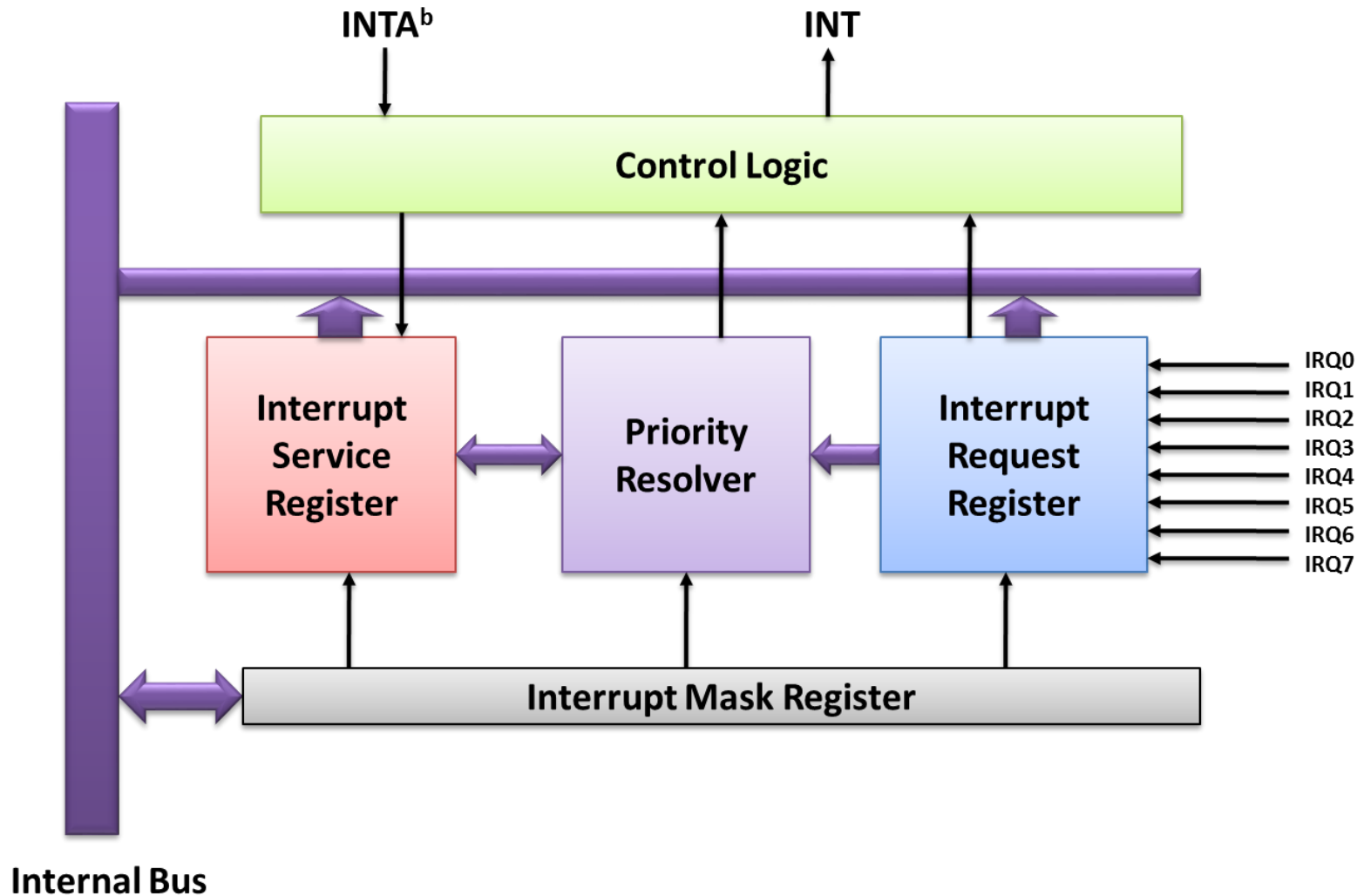
- D0-D7: Data pinleri
- IR0-IR7: Kesme istek pinleri
- $\overline{WR}$ ,  $\overline{RD}$ ,  $\overline{CS}$
- INT:  $\mu P$  INTR ucuna
- $\overline{INTA}$ :  $\mu P$   $\overline{INTA}$  ucundan
- A0: Adres ucu
- CAS2-CAS0: Kaskat seçim uçları
- $SP/\overline{EN}$ : Slave program/enable buffer



# Kesme İşlemler Sıralaması

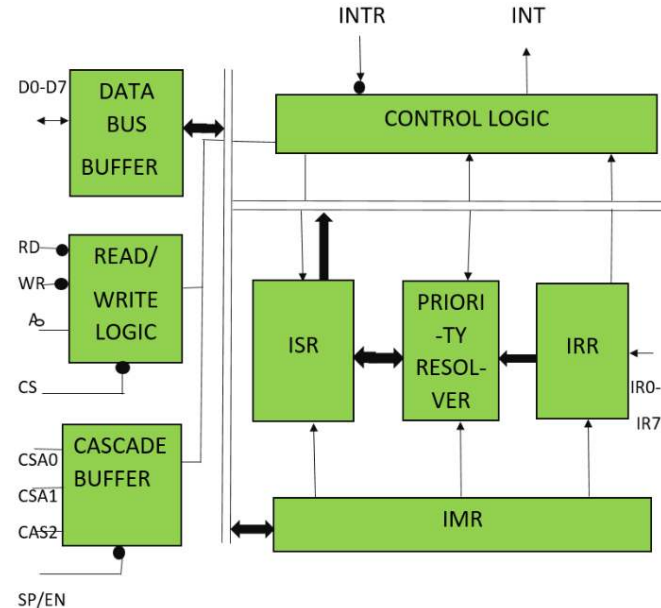
- IR7-0 uçlarından biri veya birden fazlası 1'e çıkar
- 8259 istekleri değerlendirip, CPU'ya INTR gönderir
- CPU,  $\overline{INTA}$  ile karşılık verir
- $\overline{INTA}$  8259A tarafına erişince, en yüksek öncelikli kesmeye ilişkin ISR biti 1, IRR biti 0 yapılır.
- CPU ikinci  $\overline{INTA}$  darbesini gönderir, 8259A karşılık olarak kesme vektör numarasını veriyoluna koyar.
- AEOI modunda ISR biti 0 yapılır (EOI modunda ise ISR bitini 0 yapmak için uygun bir komut beklenir.)

# 8259A İç Yapısı



## 1. Data bus buffer –

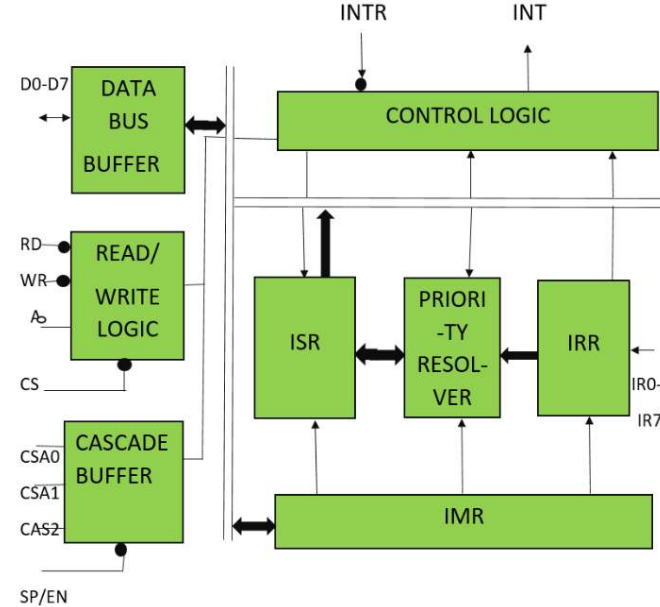
This Block is used as a mediator between 8259 and 8085/8086 microprocessor by acting as a buffer. It takes the control word from the 8085 (let say) microprocessor and transfer it to the control logic of 8259 microprocessor. Also, after selection of Interrupt by 8259 microprocessor, it transfer the opcode of the selected Interrupt and address of the Interrupt service sub routine to the other connected microprocessor. The data bus buffer consists of 8 bits represented as D0-D7 in the block diagram. Thus, shows that a maximum of 8 bits data can be transferred at a time.





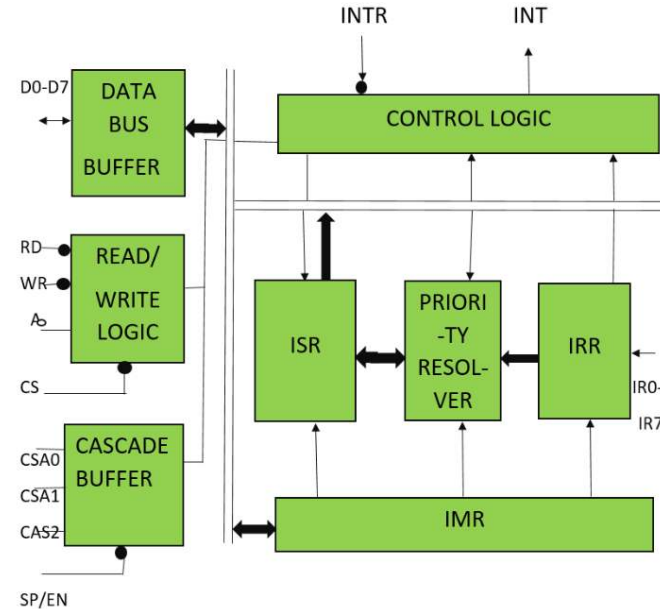
## 2. Read/Write logic –

This block works only when the value of pin CS is low (as this pin is active low). This block is responsible for the flow of data depending upon the inputs of RD and WR. These two pins are active low pins used for read and write operations.



### 3. Control logic –

It is the centre of the microprocessor and controls the functioning of every block. It has pin INTR which is connected with other microprocessor for taking interrupt request and pin INT for giving the output. If 8259 is enabled, and the other microprocessor Interrupt flag is high then this causes the value of the output INT pin high and in this way 8259 responds to the request made by other microprocessor.



#### 4. Interrupt request register (IRR) –

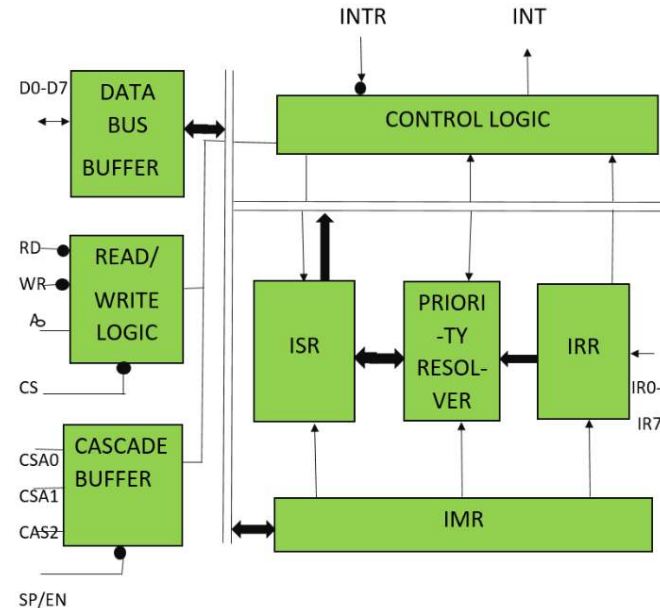
It stores all the interrupt level which are requesting for Interrupt services.

#### 5. Interrupt service register (ISR) –

It stores the interrupt level which are currently being executed.

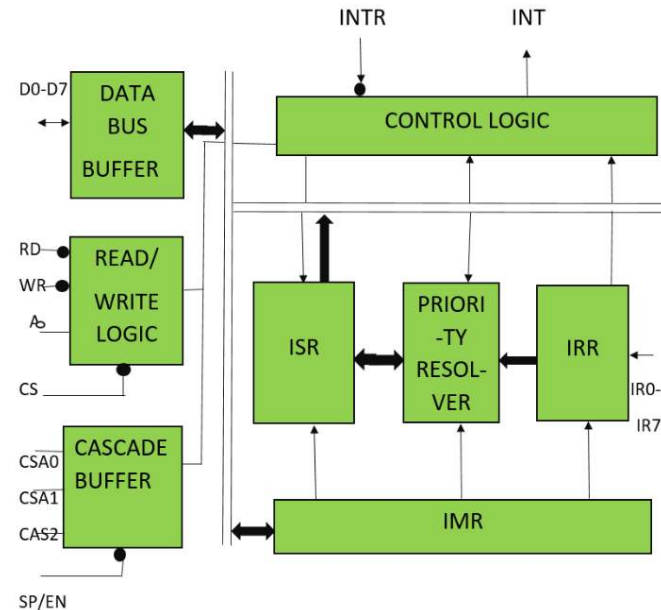
#### 6. Interrupt mask register (IMR) –

It stores the interrupt level which have to be masked by storing the masking bits of the interrupt level.



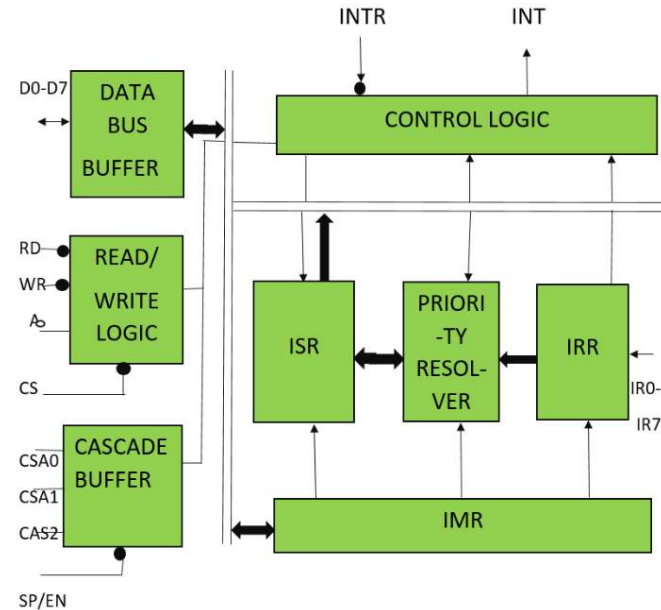
## 7. Priority resolver –

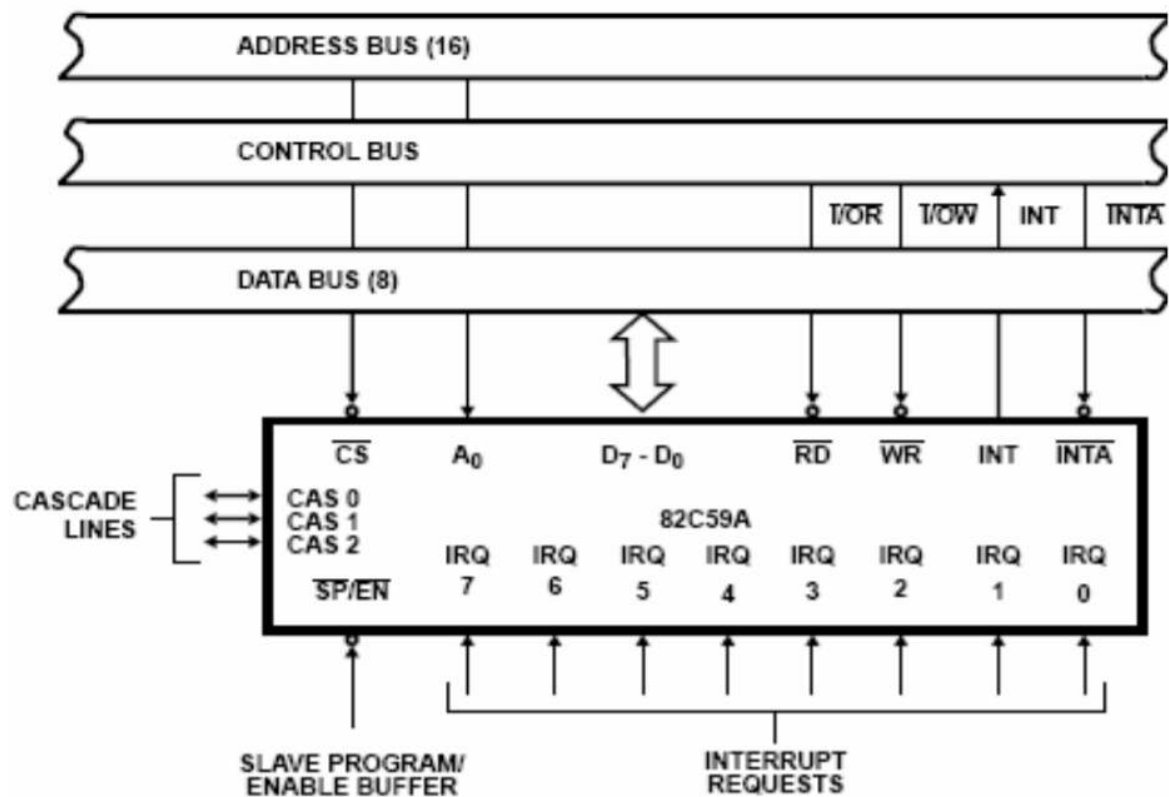
It examines all the three registers and set the priority of interrupts and according to the priority of the interrupts, interrupt with highest priority is set in ISR register. Also, it reset the interrupt level which is already been serviced in IRR.



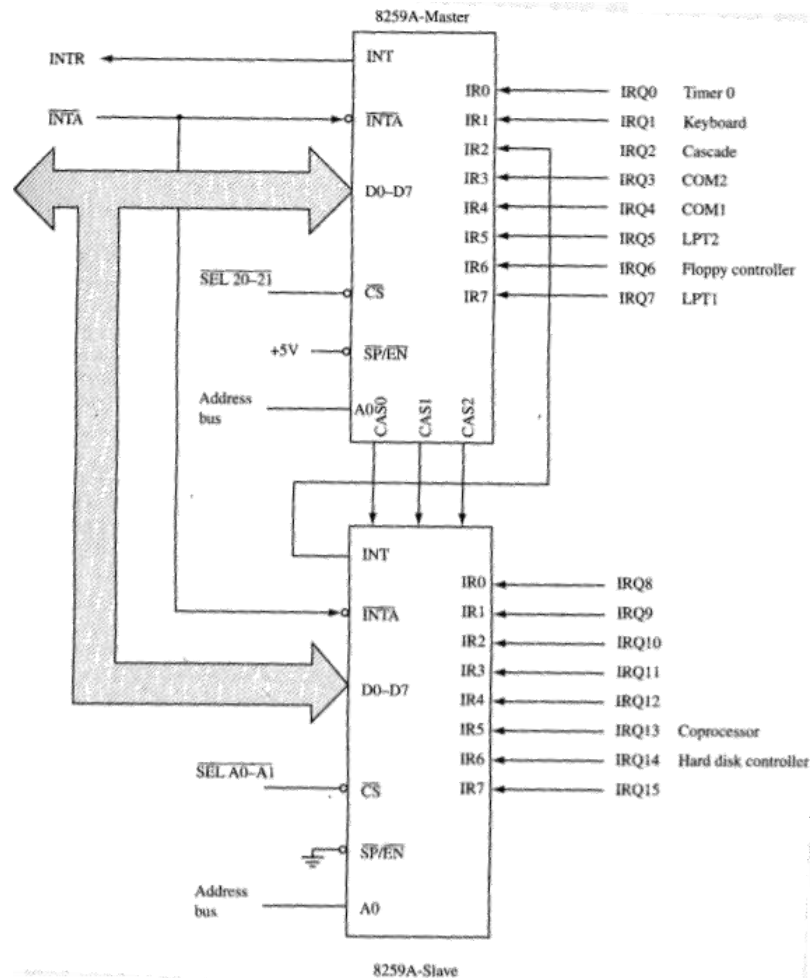
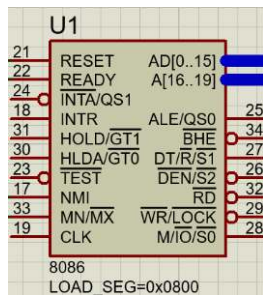
## 8. Cascade buffer –

To increase the Interrupt handling capability, we can further cascade more number of pins by using cascade buffer. So, during increment of interrupt capability, CSA lines are used to control multiple interrupt structure.





82C59A STANDARD SYSTEM BUS INTERFACE

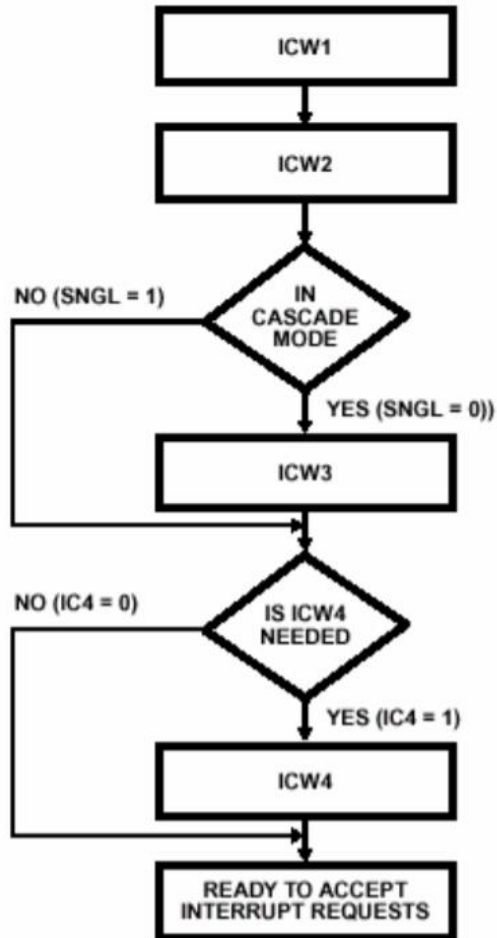


# 8259A PIC

- 8259A basit mikroişlemcili sistemlerde tek olarak kullanılacaksa  $SP/\overline{EN}$  ucu 1 olarak ayarlanır (master)
- 8259A, Initialization command words (ICWs) ve Operation command words (OCWs) kullanılarak ayarlanır

Means, in non-buffered mode, the  $\overline{SP}/\overline{EN}$  pin of an 8259 is used to specify whether the 8259 is to operate as a master or as a slave, and in the buffered mode, the  $\overline{SP}/\overline{EN}$  pin is used as an output to enable the data bus buffer of the system.





Two types of command words are provided to program the 8259:

- 1) The initialization command words (ICW)
- 2) The operational command words (OCW)

Fully Nested Mode is entered.

ICW3 and ICW4 are optional

It is not possible to modify just one ICW. Whole ICW sequence must be repeated

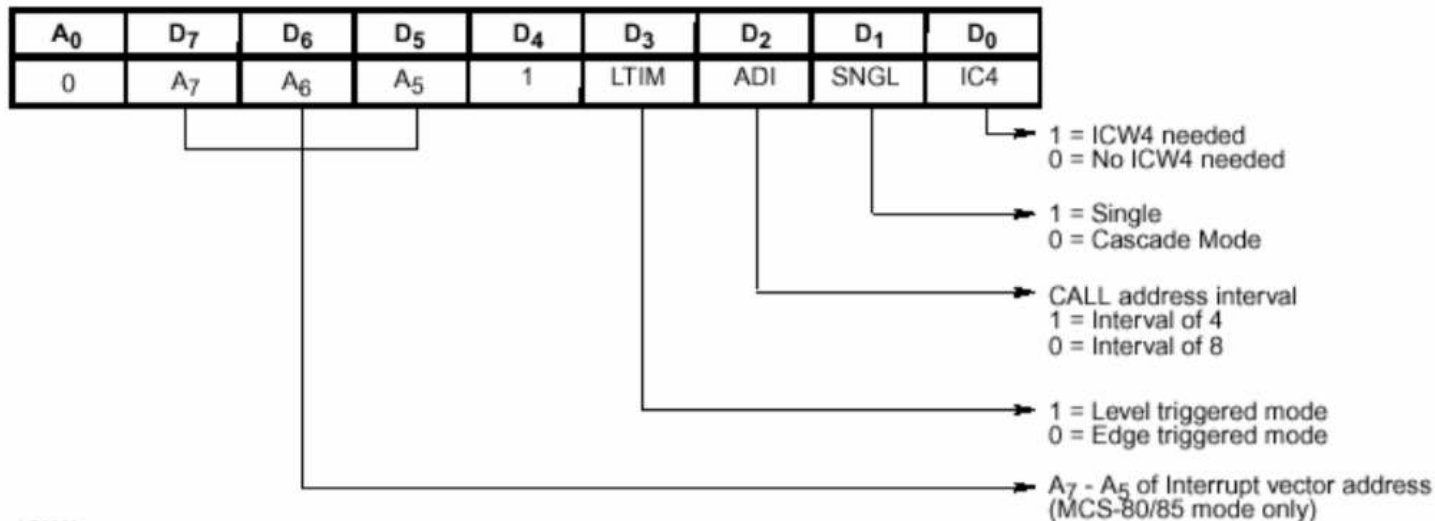
# ICW1

| AD0 | D7        | D6 | D5 | D4 | D3                                        | D2 | D1                    | D0                              |
|-----|-----------|----|----|----|-------------------------------------------|----|-----------------------|---------------------------------|
| 0   | 0         | 0  | 0  | 1  | LTIM                                      | 0  | SGNL                  | IC4                             |
|     | 0 for x86 |    |    |    | 1 for Level Trigger<br>0 for Edge Trigger |    | 1=single<br>0=Cascade | 1=IC4 needed<br>0=no IC4 needed |

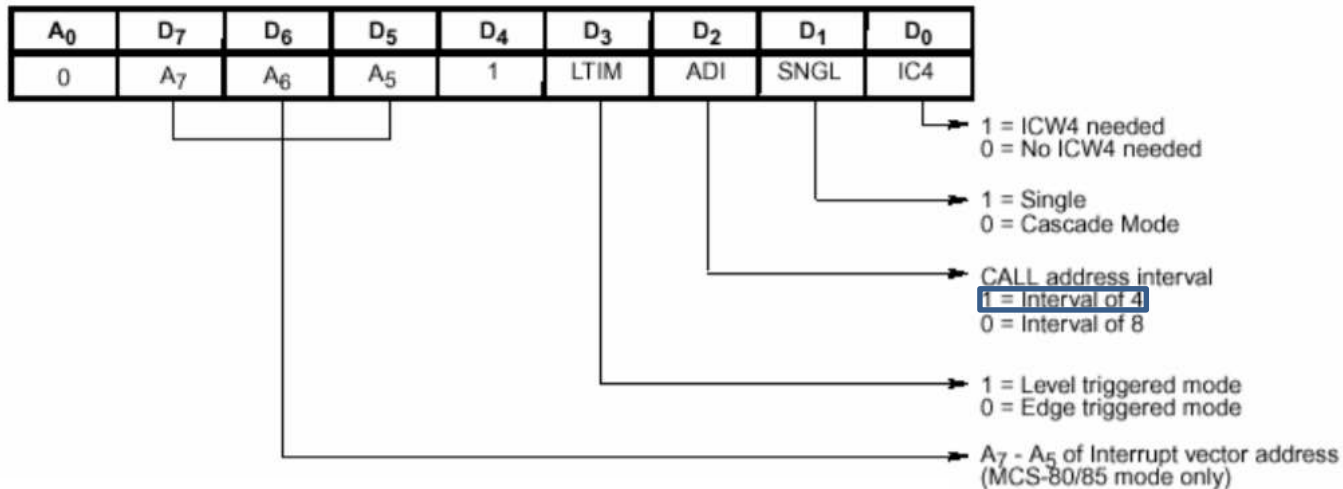
# ICW1

| AD0 | D7        | D6 | D5 | D4 | D3                                        | D2 | D1                    | D0                              |
|-----|-----------|----|----|----|-------------------------------------------|----|-----------------------|---------------------------------|
| 0   | 0         | 0  | 0  | 1  | LTIM                                      | 0  | SGNL                  | IC4                             |
|     | 0 for x86 |    |    |    | 1 for Level Trigger<br>0 for Edge Trigger |    | 1=single<br>0=Cascade | 1=IC4 needed<br>0=no IC4 needed |

ICW1



ICW1



### Initialization Command Word 1 (ICW1)

Fig. 8.8 shows the Initialization Command Word 1 (ICW1).

A write command issued to the 8259 with A<sub>0</sub> = 0 and D<sub>4</sub> = 1 is interpreted as ICW1, which starts the initialization sequence.

It specifies

1. Single or multiple 8259As in the system.
2. 4 or 8 bit interval between the interrupt vector locations.
3. The address bits A<sub>7</sub> - A<sub>5</sub> of the CALL instruction.
4. Edge triggered or level triggered interrupts.
5. ICW4 is needed or not.

Edge triggered

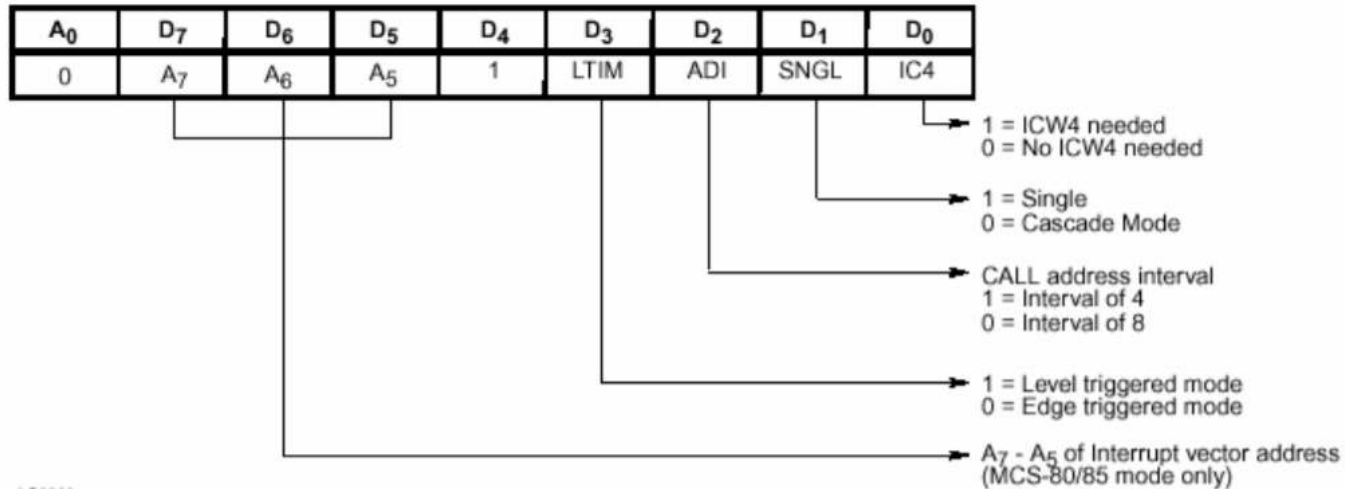


Level triggered



- Edge triggered interrupts signal as a **one shot** event!
- Level triggered interrupts are signaled **as long as line is raised**

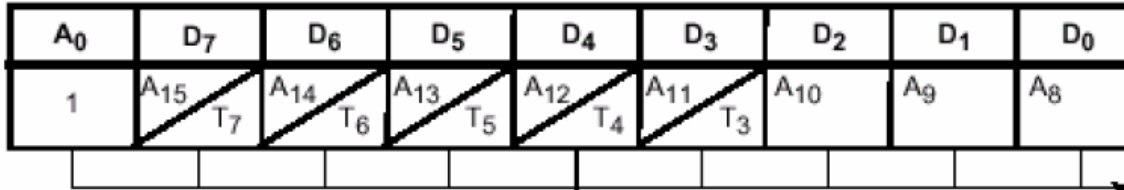
ICW1



What value should be written to ICW1 in order to configure the 8259 so that ICW4 needed, the system is going to use multiple 8259s and its inputs are level sensitive?

# ICW2

ICW2



→ A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

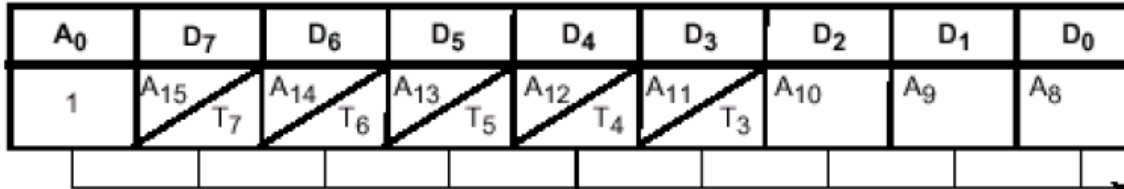
T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

| AD0 | D7                                                 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|-----|----------------------------------------------------|----|----|----|----|----|----|----|
| 1   | T7                                                 | T6 | T5 | T4 | T3 | T2 | T1 | T0 |
|     | T7=T0 is the assign to IR0, Vector address for ISR |    |    |    |    |    |    |    |

A write command following ICW1, with A<sub>0</sub> = 1 is interpreted as ICW2. This is used to load the high order byte of the interrupt vector address of all the interrupts.

# ICW2

ICW2



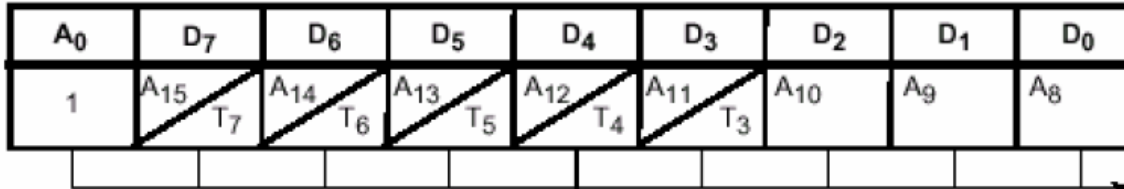
→ A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

What should be programmed into register ICW2 if type number output on the bus is to range from F0h to F7h

# ICW2

ICW2



A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

What should be programmed into register ICW2 if type number output on the bus is to range from F0h to F7h

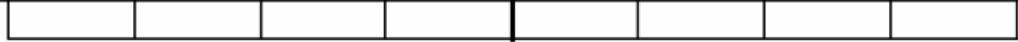
$$11110000b = F0h$$



# ICW2

ICW2

| A <sub>0</sub> | D <sub>7</sub>                    | D <sub>6</sub>                    | D <sub>5</sub>                    | D <sub>4</sub>                    | D <sub>3</sub>                    | D <sub>2</sub>  | D <sub>1</sub> | D <sub>0</sub> |
|----------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------|----------------|----------------|
| 1              | A <sub>15</sub><br>T <sub>7</sub> | A <sub>14</sub><br>T <sub>6</sub> | A <sub>13</sub><br>T <sub>5</sub> | A <sub>12</sub><br>T <sub>4</sub> | A <sub>11</sub><br>T <sub>3</sub> | A <sub>10</sub> | A <sub>9</sub> | A <sub>8</sub> |



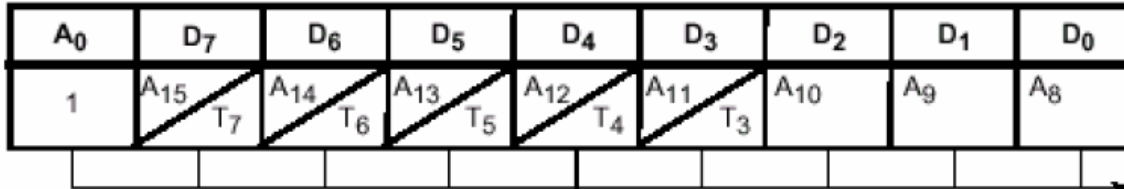
A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

Suppose IR6 is set to generate the value of 6E. Generate the addresses for the other interrupts.

# ICW2

ICW2



→ A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

Suppose IR6 is set to generate the value of 6E. Generate the addresses for the other interrupts.

$$\text{IR7} = 6\text{F}$$

$$\text{IR6} = 6\text{E}$$

$$\text{IR5} = 6\text{D}$$

$$\text{IR4} = 6\text{C}$$

$$\text{IR3} = 6\text{B}$$

$$\text{IR2} = 6\text{A}$$

$$\text{IR1} = 69$$

$$\text{IR0} = 68$$

### ICW3 (MASTER DEVICE)

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | S <sub>7</sub> | S <sub>6</sub> | S <sub>5</sub> | S <sub>4</sub> | S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> |

1 = IR input has a slave  
0 = IR input does not have a slave

### ICW3 (SLAVE DEVICE)

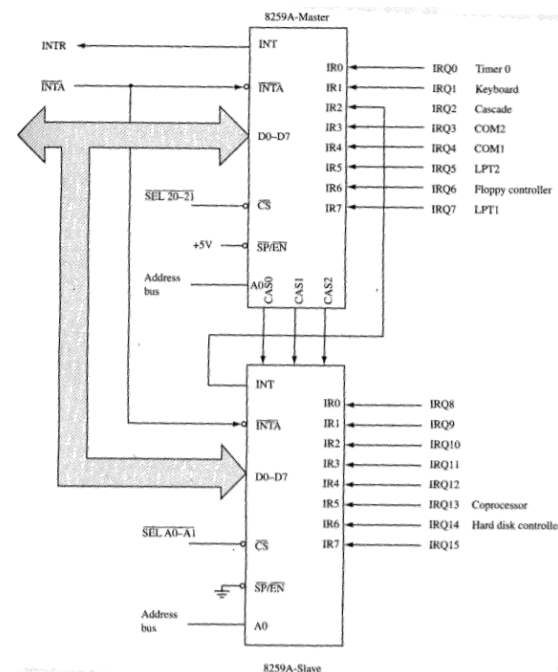
| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub>  | D <sub>1</sub>  | D <sub>0</sub>  |
|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|
| 1              | 0              | 0              | 0              | 0              | 0              | ID <sub>2</sub> | ID <sub>1</sub> | ID <sub>0</sub> |

### SLAVE ID (NOTE)

|   | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 2 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

## Initialization Command Word 3 (ICW3)

ICW3 is required only if there is more than one 8259 in the system and if they are cascaded. An ICW3 operation loads a slave register in the 8259. The format of the byte to be loaded as an ICW3 for a master 8259 or a slave is shown in the Fig. 8.10. For master, each bit in ICW3 is used to specify whether it has a slave 8259 attached to it on its corresponding IR (Interrupt Request) input. For slave, bits D<sub>0</sub>-D<sub>2</sub> of ICW3 are used to assign a slave identification code (slave ID) to the 8259.



ICW3 (MASTER DEVICE)

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | S <sub>7</sub> | S <sub>6</sub> | S <sub>5</sub> | S <sub>4</sub> | S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> |

1 = IR input has a slave  
0 = IR input does not have a slave

ICW3 (SLAVE DEVICE)

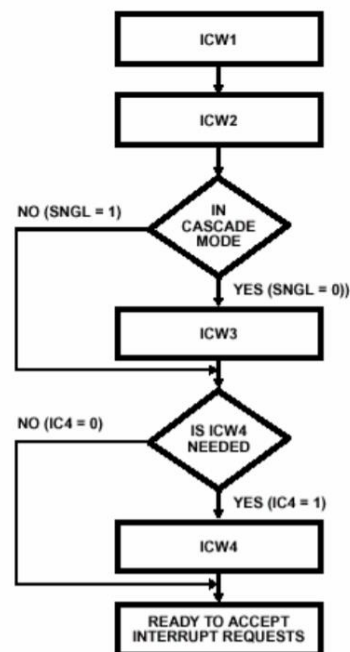
| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub>  | D <sub>1</sub>  | D <sub>0</sub>  |
|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|
| 1              | 0              | 0              | 0              | 0              | 0              | ID <sub>2</sub> | ID <sub>1</sub> | ID <sub>0</sub> |

SLAVE ID (NOTE)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

### Initialization Command Word 3 (ICW3)

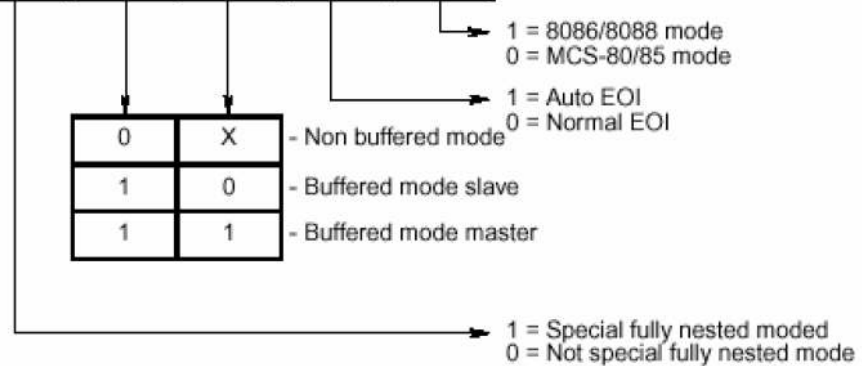
ICW3 is required only if there is more than one 8259 in the system and if they are cascaded. An ICW3 operation loads a slave register in the 8259. The format of the byte to be loaded as an ICW3 for a master 8259 or a slave is shown in the Fig. 8.10. For master, each bit in ICW3 is used to specify whether it has a slave 8259 attached to it on its corresponding IR (Interrupt Request) input. For slave, bits D<sub>0</sub>-D<sub>2</sub> of ICW3 are used to assign a slave identification code (slave ID) to the 8259.



ICW4 It is loaded only if the D<sub>0</sub> bit of ICW1 (IC 4) is set.

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | 0              | 0              | 0              | SFNM           | BUF            | M/S            | AEOL           | μPM            |

**AEOL mode** requires no commands. During the second INTA the ISR bit is reset. The major drawback with this mode is that the ISR doesn't have info on which IR is served. Thus any IR with any priority can *now* Interrupt service routine.



BUF when 1 selects buffer mode. The SP/EN pin becomes an output for the data buffers.

When 0, the SP/EN pin becomes the input for the (MASTER/SLAVE) functionality

M/S is used to set the function of the 8259 when operated in buffered mode. If M/S is set the 8259 will function as the MASTER. If cleared will function as SLAVE.

ICW4

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | 0              | 0              | 0              | SFNM           | BUF            | M/S            | AEIO           | μPM            |

It specifies.

- 1) Whether to use special fully nested mode or non special fully nested mode.
- 2) Whether to use buffered mode or non buffered mode.
- 3) Whether to use Automatic EOI or Normal EOI
- 4) CPU used, 8086/8088 or 80810.

## End of Interrupt (EOI)

The IS bit can be reset by an End of Interrupt command issued by the CPU, usually just before exiting from the interrupt routine.

## Automatic End of Interrupt (AEIOI)

If the AEIOI mode is set, the 8259 will perform a non-specific EOI on its own on the trailing edge of the third  $\overline{\text{INTA}}$  pulse. The AEIOI mode can only be used for a master 8259 and not for a slave.

### Fully nested mode:

- This is a general purpose mode where all IR's are arranged in highest to lowest.
- IR0 highest and IR7 lowest.

### Special Fully Nested Mode:

- Used in more complicated systems.
- Similar to, normal nested mode.
- When an interrupt request from a certain slave is in service, this slave can further send requests to the master.
- The master interrupts the CPU only.

### **Operation Command Word 1 (OCW1)**

A Write command to the 8259 with  $A_0 = 1$  (after ICW2) is interpreted as OCW1. OCW1 is used for enabling or disabling the recognition of specific interrupt requests by programming the IMR.

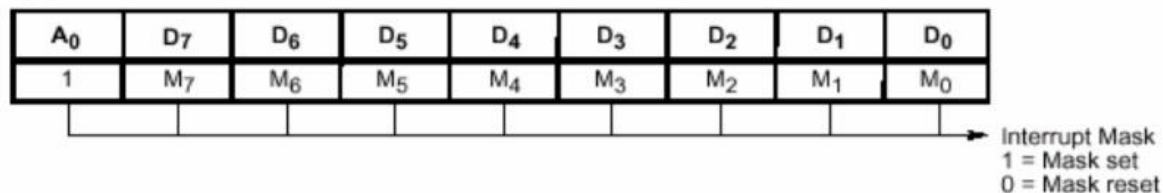
### **Operation Command Word 2 (OCW2)**

A Write command with  $A_0 = 1$  and  $D_4 D_3 = 00$  is interpreted as OCW2. The R(Rotate), SL (Select-Level), EOI bits control the Rotate and End Of Interrupt Modes and combinations of the two. Fig. 8.13 shows the Operation Command Word format.  $L_2 - L_0$  are used to specify the interrupt level to be acted upon when the SL bit is active.

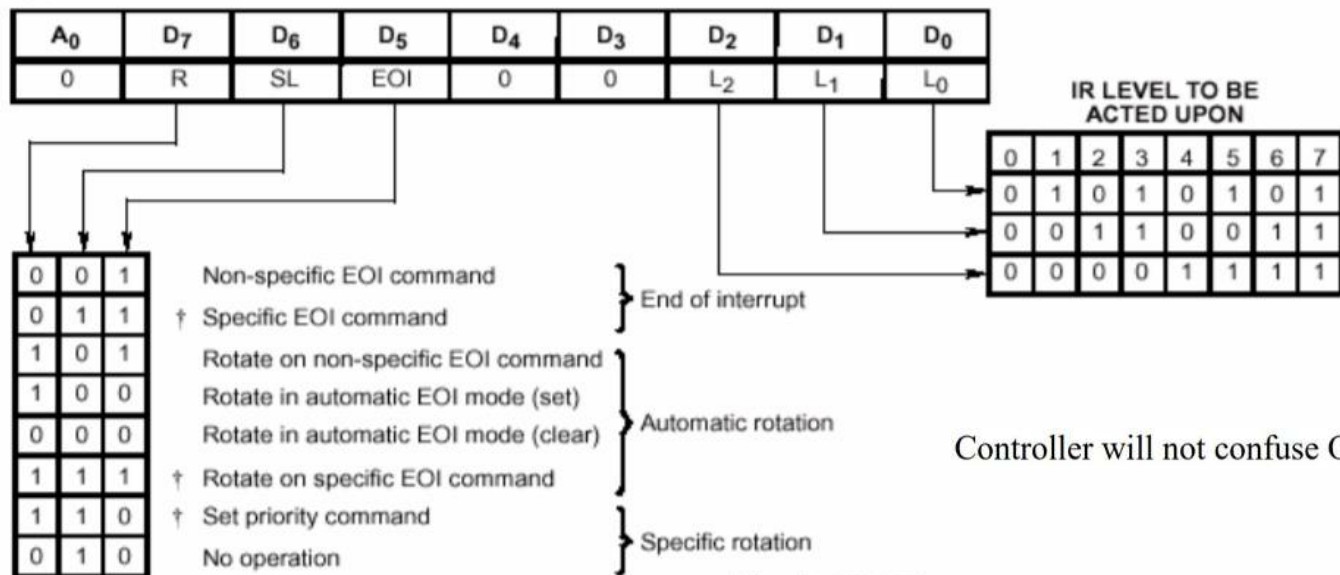


OCW1 is used to access the contents of the IMR. A READ operation can be performed to the IMR to determine the present setting of the mask. Write operations can be performed to mask or unmask certain bits.

OCW1



OCW2



Controller will not confuse OCW2 with ICW1 since D<sub>4</sub> = 1

### AUTOMATIC ROTATION MODE:

- In this mode, a device after being serviced, receives the lowest priority.

| $IR_0$ | $IR_1$ | $IR_2$ | $IR_3$ | $IR_4$ | $IR_5$ | $IR_6$ | $IR_7$ |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 4      | 5      | 6      | 7      | 0      | 1      | 2      | 3      |

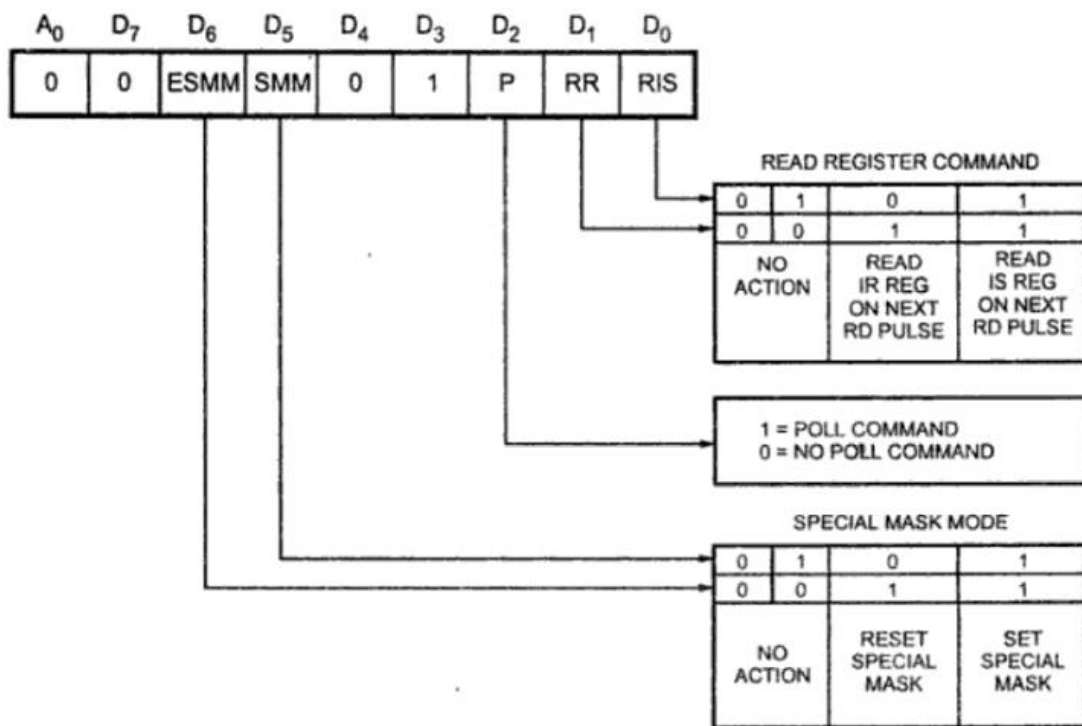
### SPECIFIC ROTATION MODE:

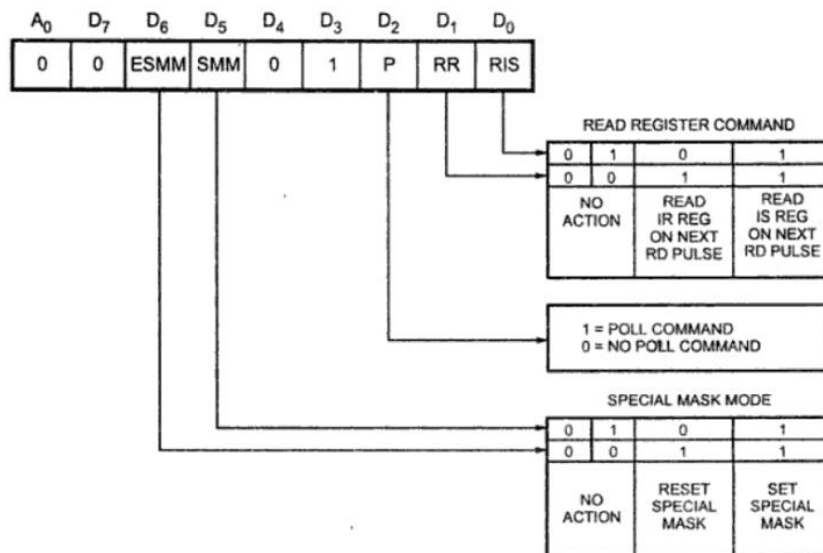
- Similar to automatic rotation mode, except that the user can select any IR for the lowest priority, thus fixing all other priorities.

| $IR_0$ | $IR_1$ | $IR_2$ | $IR_3$ | $IR_4$ | $IR_5$ | $IR_6$ | $IR_7$ |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 5      | 6      | 7      | 0      | 1      | 2      | 3      | 4      |

### Operation Command Word 3 (OCW3)

OCW3 is used to read the status of the registers, and to set or reset the Special Mask and Polled modes. Fig. 8.14 shows format operational command word 3.





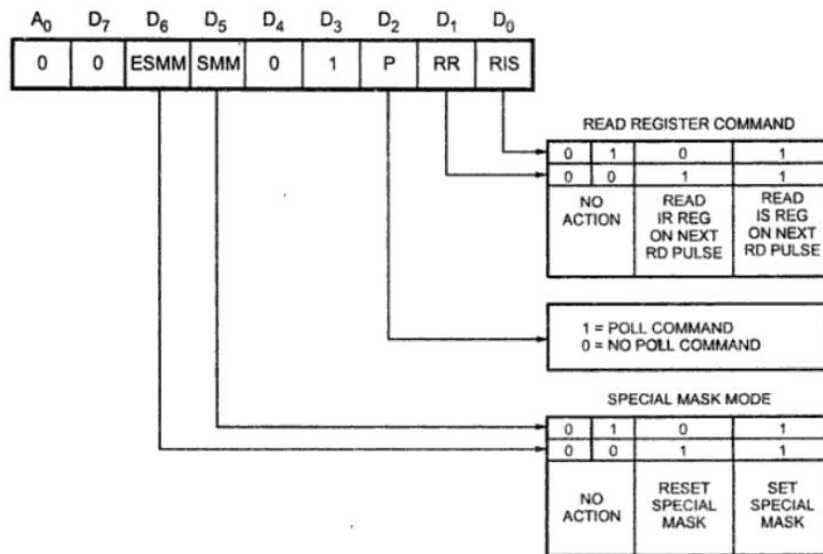
## 8259 Status Read Operations

The status of the Interrupt Request Register, the In-Service Register, and the Interrupt Mask Register of the 8259 may be read by issuing appropriate Read commands as described below.

### IRR Status Read

An OCW3 with RR (Read Register) = 1 and RIS (Read ISR) = 0 set up the 8259 for a status read of the Interrupt Request Register.

When the 8259 is not in the Polled mode, after it is set up for an IRR status read operation, all Read commands with A<sub>0</sub> = 1 cause the 8259 to send the IRR status word.

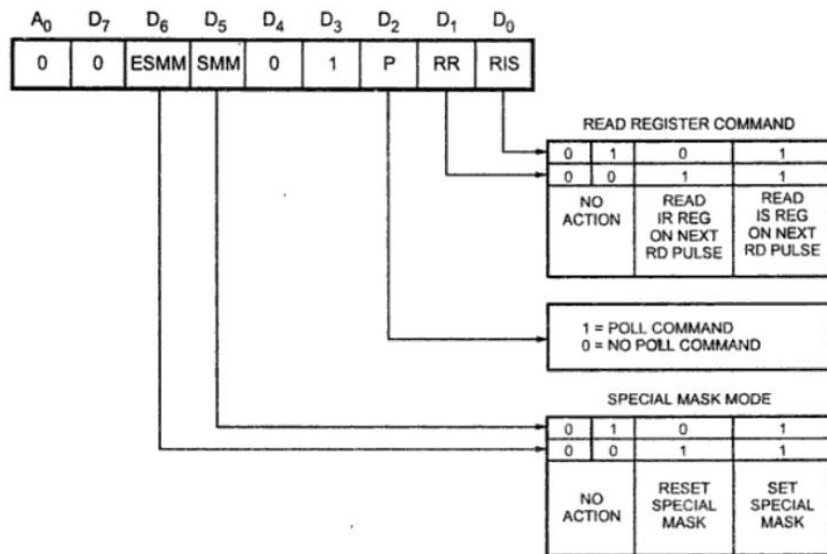


### ISR Status Read

An OCW3 with RR = 1 and RIS = 1 sets up the 8259 for a status read of the In-Service Register. A subsequent read command issued to the 8259 will cause the 8259 to send the contents of the ISR onto the data bus.

### IMR Status Read

A Read command issued to the 8259 with A<sub>0</sub> = 1 (with  $\overline{RD}$ ,  $\overline{CS}$  = 0) causes the 8259 to put out the contents of the Interrupt Mask Register. OCW3 is not required for a status read of the IMR.



#### e) Poll Mode :

In this mode the INT output is not used. The microprocessor checks the status of interrupt requests by issuing poll command. The microprocessor reads contents of 8259A after issuing poll command. During this read operation the 8259A provides polled word and sets ISR bit of highest priority active interrupt request FORMAT.

|   |   |   |   |   |                |                |                |
|---|---|---|---|---|----------------|----------------|----------------|
| I | X | X | X | X | W <sub>2</sub> | W <sub>1</sub> | W <sub>0</sub> |
|---|---|---|---|---|----------------|----------------|----------------|

I = 1 → One or more interrupt requests activated.

I = 0 → No interrupt request activated.

W<sub>2</sub> W<sub>1</sub> W<sub>0</sub> → Binary code of highest priority active interrupt request.

# OCW1 & OCW2 & OCW3

| AD0 | D7                                                  | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|-----|-----------------------------------------------------|----|----|----|----|----|----|----|
| 1   | M7                                                  | M6 | M5 | M4 | M3 | M2 | M1 | M0 |
|     | <b>Mi: Interrupt mask, 1=mask set, 0=mask reset</b> |    |    |    |    |    |    |    |

| AD0 | D7     | D6       | D5  | D4 | D3 | D2                              | D1 | D0 |
|-----|--------|----------|-----|----|----|---------------------------------|----|----|
| 0   | R      | SL       | EOI | 0  | 0  | L2                              | L1 | L0 |
|     | Rotate | Specific | EOI |    |    | IR Level to be acted Upon (0-7) |    |    |

| AD0 | D7 | D6                | D5  | D4 | D3 | D2           | D1                 | D0  |
|-----|----|-------------------|-----|----|----|--------------|--------------------|-----|
| 0   | 0  | ESMM              | SMM | 0  | 1  | P            | RR                 | RIS |
|     |    | Special Mask Mode |     |    |    | Poll Command | Register Read Mode |     |

# 8259A Ayarlama – ICW'ler

| CS | A0 | Initialization |
|----|----|----------------|
| 0  | 0  | ICW1           |
| 0  | 1  | ICW2,ICW3,ICW4 |
| 1  | X  | Not Address    |



# 8259A Ayarlama – OCW'ler

| CS | A0 | Operation<br>Command Word |
|----|----|---------------------------|
| 0  | 0  | OCW2, OCW3                |
| 0  | 1  | OCW1                      |
| 1  | X  | Not Address               |

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

2. The register that stores all the interrupt requests in it in order to serve them one by one on a priority basis is
- a) Interrupt Request Register
  - b) In-Service Register
  - c) Priority resolver
  - d) Interrupt Mask Register

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

2. The register that stores all the interrupt requests in it in order to serve them one by one on a priority basis is

- a) Interrupt Request Register
- b) In-Service Register
- c) Priority resolver
- d) Interrupt Mask Register

 [View Answer](#)

Answer: a

Explanation: The interrupts at IRQ input lines are handled by Interrupt Request Register internally.

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

3. The register that stores the bits required to mask the interrupt inputs is
- a) In-service register
  - b) Priority resolver
  - c) Interrupt Mask register
  - d) None

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

3. The register that stores the bits required to mask the interrupt inputs is

- a) In-service register
- b) Priority resolver
- c) Interrupt Mask register
- d) None

 View Answer

Answer: c

Explanation: Also, Interrupt Mask Register operates on IRR(Interrupt Request Register) at the direction of the Priority Resolver.

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

4. The interrupt control logic

- a) manages interrupts
- b) manages interrupt acknowledge signals
- c) accepts interrupt acknowledge signal
- d) all of the mentioned

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

4. The interrupt control logic

- a) manages interrupts
- b) manages interrupt acknowledge signals
- c) accepts interrupt acknowledge signal
- d) all of the mentioned

 View Answer

Answer: d

Explanation: The interrupt control logic performs all the operations that are involved within the interrupts like accepting and managing interrupt acknowledge signals, interrupts.

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

5. In a cascaded mode, the number of vectored interrupts provided by 8259A is

- a) 4
- b) 8
- c) 16
- d) 64



## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

5. In a cascaded mode, the number of vectored interrupts provided by 8259A is

- a) 4
- b) 8
- c) 16
- d) 64

 View Answer

Answer: d

Explanation: A single 8259A provides 8 vectored interrupts. In cascade mode, 64 vectored interrupts can be provided.

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

6. When the PS(active low)/EN(active low) pin of 8259A used in buffered mode, then it can be used as a
- a) input to designate chip is master or slave
  - b) buffer enable
  - c) buffer disable
  - d) none

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

6. When the PS(active low)/EN(active low) pin of 8259A used in buffered mode, then it can be used as a
- a) input to designate chip is master or slave
  - b) buffer enable
  - c) buffer disable
  - d) none

 View Answer

Answer: b

Explanation: When the pin is used in buffered mode, then it can be used as a buffer enable to control buffer transreceivers. If it is not used in buffered mode, then the pin is used as input to designate whether the chip is used as a master or a slave.

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

9. In the application where all the interrupting devices are of equal priority, the mode used is
- a) Automatic rotation
  - b) Automatic EOI mode
  - c) Specific rotation
  - d) EOI

## Microprocessors Questions and Answers – Programmable Interrupt Controller 8259A

9. In the application where all the interrupting devices are of equal priority, the mode used is

- a) Automatic rotation
- b) Automatic EOI mode
- c) Specific rotation
- d) EOI

 View Answer

Answer: a

Explanation: The automatic rotation is used in the applications where all the interrupting devices are of equal priority.

60H Adresinden itibaren ardışık çift adreslere 8259 yerleştiriliyor.  
8259'un IRO ucuna pull-down dirençli bir buton bağlı.

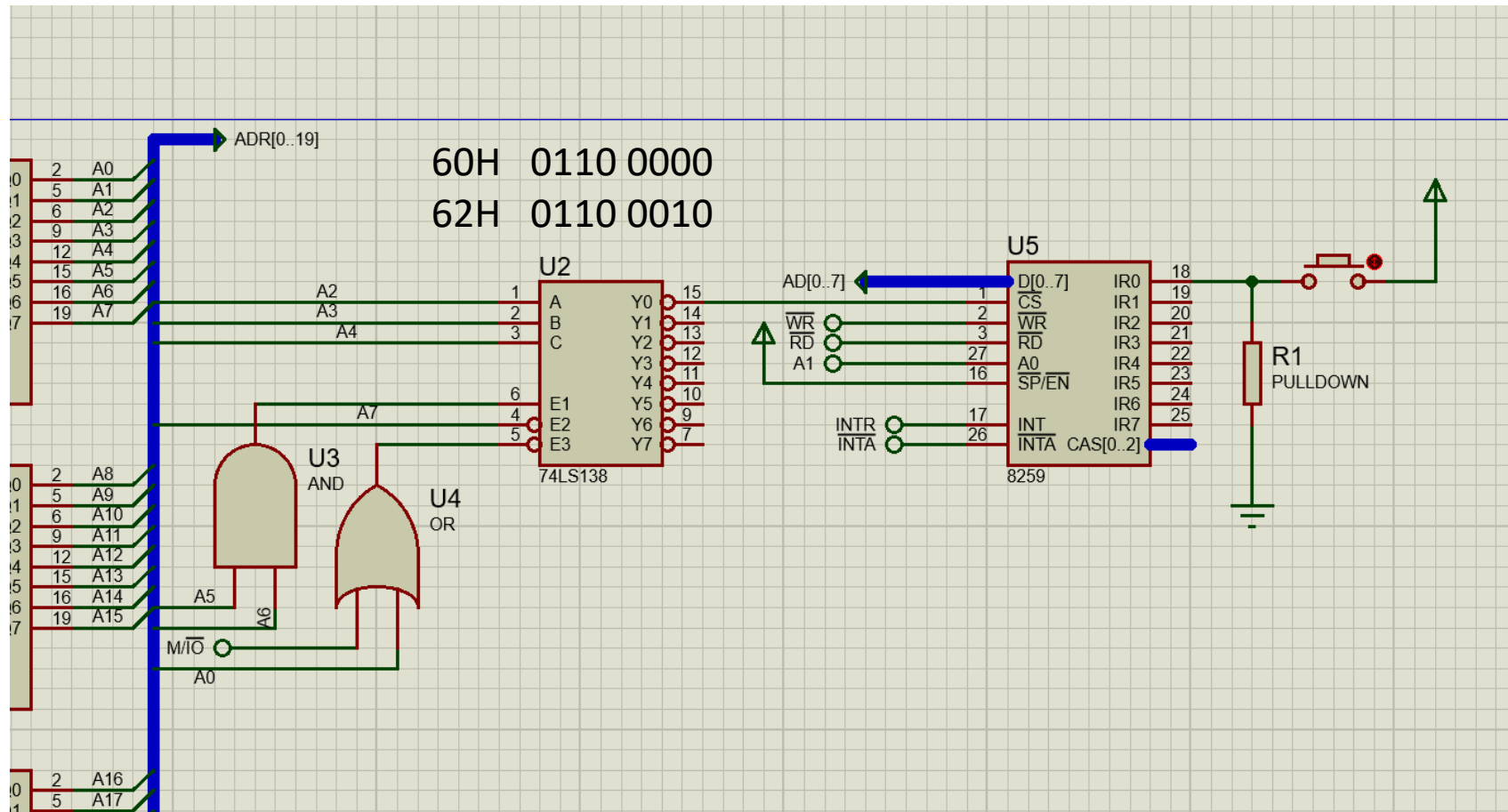
- 1.) Butona basıldığında 40H tipinde kesme tetiklenmeli.
- 2.) 40H tipindeki kesme ile butona kaç kez basıldığı sayılmalı.

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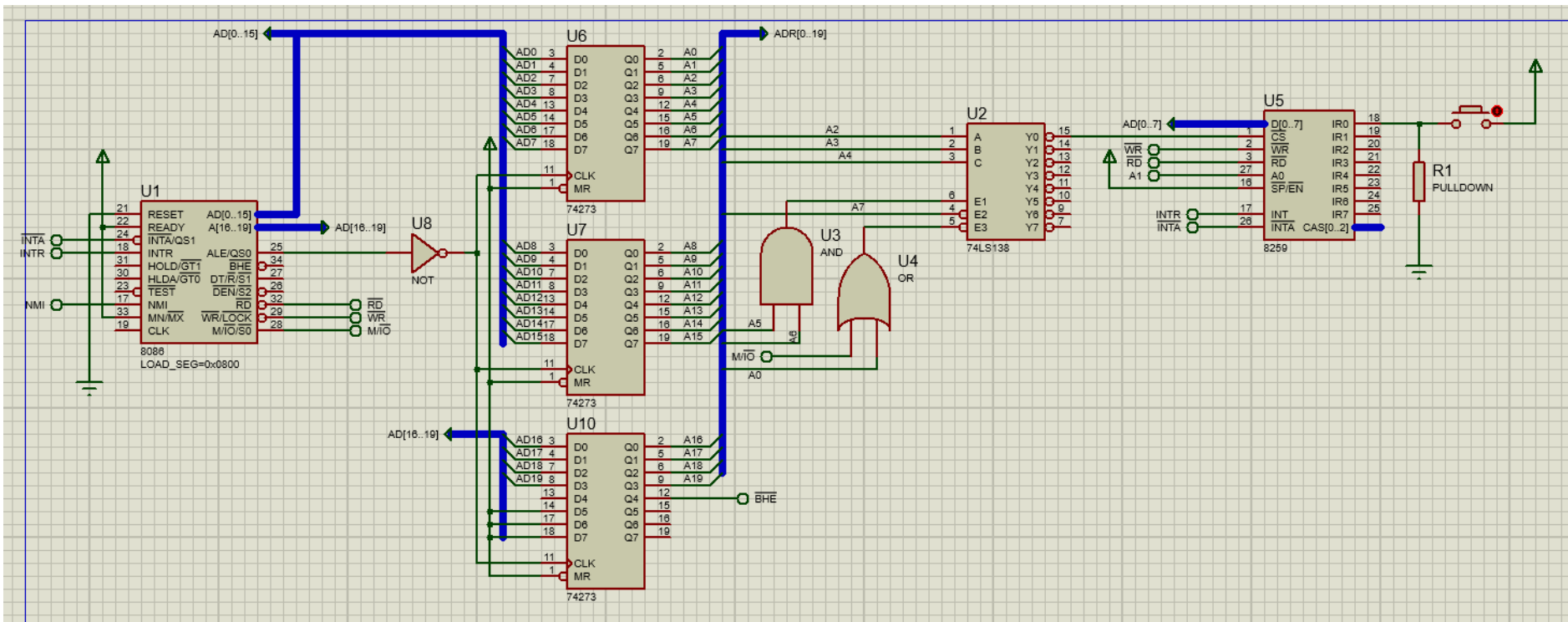
- 1.) Butona basıldığında 40H tipinde kesme tetiklenmeli.
- 2.) 40H tipindeki kesme ile butona kaç kez basıldığı sayılmalı.

60H 0110 0000

62H 0110 0010







```

49 MOV AL, 13H
50 OUT 60H, AL
51 MOV AL, 40H
52 OUT 62H, AL
53 MOV AL, 03H
54 OUT 62H, AL
55 STI
56 XOR AX, AX
57 ENDLESS:

```

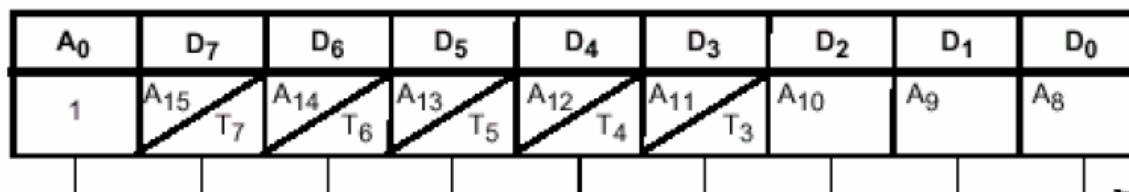
0001 0011

ICW1

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 0              | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | 1              | LTIM           | ADI            | SNGL           | IC4            |

- 1 = ICW4 needed  
0 = No ICW4 needed
- 1 = Single  
0 = Cascade Mode
- CALL address interval  
1 = Interval of 4  
0 = Interval of 8
- 1 = Level triggered mode  
0 = Edge triggered mode
- A<sub>7</sub> - A<sub>5</sub> of Interrupt vector address  
(MCS-80/85 mode only)

ICW2



→ A<sub>15</sub> - A<sub>8</sub> of interrupt vector address  
(MCS80/85 mode)

T<sub>7</sub> - T<sub>3</sub> of interrupt vector address  
(8086/8088 mode)

```

49 MOV AL,
50 OUT 60H, AL
51 MOV AL, 40H
52 OUT 62H, AL
53 MOV AL, 03H
54 OUT 62H, AL
55 STI
56 XOR AX, AX
57 ENDLESS:

```

0100 0000

```

49 MOV AL, 13H
50 OUT 60H, AL
51 MOV AL, 40H
52 OUT 62H, AL
53 MOV AL, 03H
54 OUT 62H, AL
55 STI
56 XOR AX, AX
57 ENDLESS:

```

0000 0011

**AEOI mode** requires no commands. During the second INTA the ISR bit is reset. The major drawback with this mode is that the ISR doesn't have info on which IR is served. Thus any IR with any priority can *now* Interrupt service routine.

ICW4

| A <sub>0</sub> | D <sub>7</sub> | D <sub>6</sub> | D <sub>5</sub> | D <sub>4</sub> | D <sub>3</sub> | D <sub>2</sub> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | 0              | 0              | 0              | SFNM           | BUF            | M/S            | AEOI           | μPM            |

|   |   |
|---|---|
| 0 | X |
| 1 | 0 |
| 1 | 1 |

- 1 = 8086/8088 mode  
0 = MCS-80/85 mode
- 1 = Auto EOI  
0 = Normal EOI
- 0 - Non buffered mode
- 1 0 - Buffered mode slave
- 1 1 - Buffered mode master
- 1 = Special fully nested moded  
0 = Not special fully nested mode

BUF when 1 selects buffer mode. The SP/EN pin becomes an output for the data buffers.

When 0, the SP/EN pin becomes the input for the (MASTER/SLAVE) functionality

M/S is used to set the function of the 8259 when operated in buffered mode. If M/S is set the 8259 will function as the MASTER. If cleared will function as SLAVE.

```

10  ;=====
11  STAK    SEGMENT PARA STACK 'STACK'
12          DW 20 DUP(?)
13  STAK    ENDS
14
15  DATA    SEGMENT PARA 'DATA'
16  MYDAT    DB 20 DUP(0)
17  DATA    ENDS
18
19  CODE     SEGMENT PARA 'CODE'
20          ASSUME CS:CODE, DS:DATA, SS:STAK
21
22  NEWINT   PROC FAR
23          PUSH BP
24          MOV BP, SP
25          INC AX
26
27          POP BP
28          IRET
29  NEWINT   ENDP
30
31  START   PROC FAR
32          MOV AX, DATA
33          MOV DS, AX
34
35
36          XOR AX, AX
37          MOV ES, AX
38          MOV AL, 40H
39          MOV AH, 4
40          MUL AH
41          MOV BX, AX
42          LEA AX, NEWINT
43          MOV WORD PTR ES:[BX], AX
44          MOV AX, CS
45          MOV WORD PTR ES:[BX+2], AX
46
47
48
49  MOV AL, 13H
50  OUT 60H, AL
51  MOV AL, 40H
52  OUT 62H, AL
53  MOV AL, 03H
54  OUT 62H, AL
55  STI
56  XOR AX, AX
57  ENDLESS:
58
59
60
61  JMP ENDLESS
62  RET
63  START ENDP

```

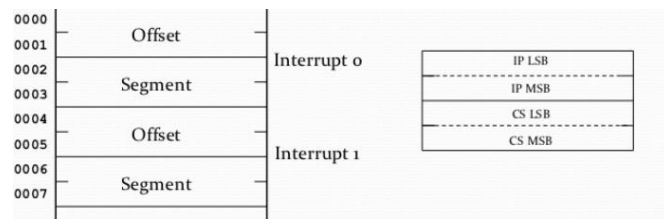
```

19 CODE      SEGMENT PARA 'CODE'
20          ASSUME CS:CODE, DS:DATA, SS:STAK
21
22 NEWINT     PROC FAR
23          PUSH BP
24          MOV BP, SP
25          INC AX
26
27          POP BP
28          IRET
29 NEWINT     ENDP
30
31 START     PROC FAR
32          MOV AX, DATA
33          MOV DS, AX
34
35
36          XOR AX, AX
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38          MOV AL, 40H
39          MOV AH, 4
40          MUL AH
41          MOV BX, AX
42          LEA AX, NEWINT
43          MOV WORD PTR ES:[BX], AX
44          MOV AX, CS
45          MOV WORD PTR ES:[BX+2], AX
46

```

## Base pointer and local variables

The base pointer is conventionally used to mark the start of a function's stack frame,



Given a Vector, where is the  
ISR address stored in memory?

$$Offset = Type \times 4$$

- LEA means Load Effective Address
- MOV means Load Value

In short, LEA loads a pointer to the item you're addressing whereas MOV loads the actual value at that address.

The purpose of LEA is to allow one to perform a non-trivial address calculation and store the result [for later usage]

```
LEA ax, [BP+SI+5] ; Compute address of value
```

```
MOV ax, [BP+SI+5] ; Load value at that address
```

```
19 CODE      SEGMENT PARA 'CODE'
20           ASSUME CS:CODE, DS:DATA, SS:STAK
21
22 NEWINT     PROC FAR
23           PUSH BP
24           MOV BP, SP
25           INC AX
26
27           POP BP
28           IRET
29 NEWINT     ENDP
30
31 START     PROC FAR
32           MOV AX, DATA
33           MOV DS, AX
34
35
36           XOR AX, AX
37           MOV ES, AX
38           MOV AL, 40H
39           MOV AH, 4
40           MUL AH
41           MOV BX, AX
42           LEA AX, NEWINT
43           MOV WORD PTR ES:[BX], AX
44           MOV AX, CS
45           MOV WORD PTR ES:[BX+2], AX
46
```

## Review Questions

1. What do you mean by interrupt ?
2. What is interrupt service routine ?
3. What are the sources of interrupts in 8086 ?
4. What is interrupt vector table ?
5. Draw and explain the IVT for 8086.
6. Briefly describe the conditions which cause the 8086 to perform each of the following types of interrupts : Type 0, Type 1, Type 2, Type 3 and Type 4.
7. Explain interrupt structure of 8086.
8. What are software interrupt ? How 8086 responds to software interrupts ?
9. Draw and explain the interrupt acknowledge cycle of 8086.
10. Describe the response of 8086 to the interrupt coming on pin.
11. What do you mean by interrupt priorities ?
12. State the interrupt priorities for 8086 interrupts.
13. What are advantages of using 8259 ?
14. List the features of 8259.
15. Explain the operating modes of 8259.
16. Draw and explain the interfacing of 8259 with 8086.
17. Draw and explain the interfacing of cascaded 8259s with 8086.
18. Explain the procedure of interrupt programming.